

NIKHIL SAHU

Cybersecurity and Digital Forensics | Final Year B.Tech
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PROFESSIONAL SUMMARY

Final-year Cybersecurity and Digital Forensics student with 3+ years of bug bounty hunting on HackerOne across 10+ real-world programs, including a responsibly disclosed vulnerability in Quick Heal's production system. Led a 4-member penetration testing team at Codevirus Security, delivering CVSS-scored VAPT reports for clients. Developed an AI-based zero-day malware detection framework and holds an Indian Patent. Seeking opportunities to apply offensive security skills in web application pentesting, VAPT, and malware analysis.

PROFESSIONAL EXPERIENCE

Penetration Testing Intern - Team Lead | Codevirus Security

January 2026 - March 2026 | Remote

- Led a 4-member intern team conducting VAPT on 3+ client web applications; managed task delegation, daily standups, and final report delivery.
- Performed black-box and grey-box assessments, identifying vulnerabilities including SQL Injection, XSS, IDOR, and broken authentication.
- Documented findings using CVSS v3.1 severity scoring and provided remediation recommendations adopted by clients.

Cyber Security Intern | Samatrix Consulting Pvt. Ltd.

August 2025 | Indore

- Conducted malware analysis in a controlled sandbox environment, identifying malicious behavior patterns and classifying threat types.
- Assisted in designing cybersecurity defense mechanisms and contributed to threat detection activities for client-facing security projects.

Security Researcher - Bug Bounty Hunter | HackerOne

2021 - 2024 | Remote

- Participated in 10+ bug bounty programs over 3+ years, identifying and reporting authentication bypasses, misconfigurations, and insecure API implementations across live production systems.
- Performed full-cycle reconnaissance, endpoint enumeration, and manual exploitation using Burp Suite and Linux-based tools.
- Demonstrated professional responsible disclosure practices - coordinating directly with security teams to verify and remediate findings.
- Identified and responsibly disclosed a brute force vulnerability in Quick Heal's product key validation system; vulnerability was acknowledged and patched following disclosure.

Cyber Security Trainee | AlanceSec

April 2024 - July 2024 | Remote

- Completed intensive hands-on training in vulnerability assessment, exploitation techniques, and penetration testing methodologies.
- Performed web application testing using Burp Suite, Metasploit, and Nmap, identifying and exploiting common OWASP Top 10 vulnerabilities.
- Practiced attack-defense simulations in controlled cybersecurity lab environments.

EDUCATION

B.Tech - Computer Technology (Cyber Security and Digital Forensics)

SAGE University Indore | CGPA: 7.82 / 10 (up to 6th Semester) | 2023 - 2027 | Indore, India

TECHNICAL SKILLS

Offensive Security: Penetration Testing, VAPT, Bug Bounty Hunting, Reconnaissance, Vulnerability Research, Exploit Development

Web Application Security: OWASP Top 10, SQL Injection, XSS, IDOR, Authentication Bypass, Session Analysis

Security Tools: Burp Suite, Nmap, Metasploit, Wireshark, Nessus, Kali Linux

Malware Analysis: Dynamic Analysis, Behavior Pattern Detection, Sandboxing

Development: Python, JavaScript, HTML, CSS, SQL, Bash Scripting, Flask

Systems and Networking: Linux, Command Line, TCP/IP, Network Protocols, Packet Analysis

Languages: Hindi (Native), English (Professional Working Proficiency)

PROJECTS

Windows Persistence and Red Team Simulation | *Reverse Shell, Windows Shell Scripting*

Simulated real-world red team operations in a controlled lab - implemented Windows persistence techniques, reverse shell mechanisms, and registry-based startup methods.

Cryptographic Hash Generator | *Python, Cryptography*

Python-based hash generator supporting MD5, SHA-256, SHA3-224, BLAKE2s, and SHAKE-128 for file integrity verification.

GitHub: github.com/heynick1337/hashgen

Web Authentication System | *Python, Flask, HTML, CSS, JavaScript, MySQL*

Built a secure web authentication system with user validation, session management, and access control following OWASP principles.

GitHub: github.com/heynick1337/AuthShield

Bug Bounty Recon Framework | *Python, Flask, HTML, CSS, JavaScript*

Web-based recon dashboard for bug bounty hunting. Features subdomain enumeration, port scanning, security header analysis, and tech stack detection with report export.

GitHub: github.com/heynick1337/recon-framework

PATENTS AND RESEARCH WORK

Adaptive AI-Based Zero-Day Malware Detection and Neutralization Device | *February 2026*

SAGE University Indore - Designed an AI-driven cybersecurity framework for real-time zero-day malware detection, autonomous threat response, and adaptive protection across enterprise and IoT environments.

System and Method for AI-Based Community Weather Forecasting | *Indian Patent* | *February 2025*

SAGE University Indore - Developed an AI-based localized weather forecasting system using machine learning algorithms. Filed and published through the Indian Patent Office.

CERTIFICATIONS

Master's in Cybersecurity and Penetration Testing (Certification) | *Jetking Indore* | *AlanceSec Mumbai*

Completed an intensive hands-on program covering ethical hacking, penetration testing, vulnerability assessment, and web application security using Burp Suite, Metasploit, Nmap, and Kali Linux across attack-defense lab simulations aligned with OWASP Top 10 standards.